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Exploration of self- and world-experiences in depersonalization traits

Anna Ciaunica[a,b,c], Elizabeth Pienkos[d], Estelle Nakul[e], Luis Madeira[f] and Harry Farmer[g]

[a]Institute of Philosophy, University of Porto, via Panoramica s/n 4150-564 Porto, Portugal; [b]Institute of Cognitive Neuroscience, University College London, WC1N 3AR London, UK; [c]Centre for Philosophy of Science, University of Lisbon, Campo Grande, 1749-016 Lisbon, Portugal; [d]Department of Psychology, Clarkson University, Potsdam, NY, USA; [e]Aix Marseille Univ, CNRS, LNC, FR3C, Marseille, France; [f]Faculdade de Medicina, Universidade de Lisboa, Lisboa, Portugal; [g]School of Human Sciences, University of Greenwich, UK

ABSTRACT

This paper proposes a qualitative study exploring anomalous self and world-experiences in individuals with high levels of depersonalization experiences. Depersonalization (DP) is a condition characterized by distressing feelings of being a detached, neutral and disembodied onlooker of one's mental and bodily processes. Our findings indicate the presence of a wide range of anomalous experiences traditionally understood to be core features of DP, such as disembodiment and disrupted self-awareness. However, our results also indicate experiential features that are less highlighted in previous work, such as faster time perception and blurriness of the self/other boundaries which may play a key role in altering one's sense of self and sense of presence in the world. Our qualitative study provides an in-depth examination of self-reported disturbances of one's relatedness to one's self and the world, thereby shedding further light on the nature of altered subjective experiences in DP. In doing so, this paper draws attention to key aspects yet overlooked that may prove valuable for shedding further light into the phenomenon of depersonalization. We conclude by highlighting limitations of this study and a number of open questions that further work needs to address in the future.

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1 Introduction

In daily life, our experiences seem to be tacitly infused by a sense of self – the feeling that there is a "real me" or "self" intimately linked to my body as the subject of sensations, experiences and thoughts, and which is distinct from the rest of the world and others. In addition, we also seem to experience a sense of presence or "realness" – the feeling that there is a "real world" out there, with which I can fully engage, exchange and interact.

CONTACT Anna Ciaunica amaciaunica@fc.ul.pt Centre for Philosophy of Science, University of Lisbon, Campo Grande, 1749-016 Lisbon, Portugal

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A significant body of work across disciplines and theoretical frameworks outlined that our sense of self is a fundamental aspect of our conscious experiences (Merleau-Ponty, 1945; Gallagher, 2000; Damasio 2000; L. A. Sass & Parnas, 2003; Zahavi, 2005; Blanke & Metzinger, 2009; Hohwy, 2007; Ciaunica & Fotopoulou, 2017; Friston, 2017; see also, Qin et al., 2020 for a recent review). But what if instead of being the first-person subject of my perceptions, emotions and actions, I constantly endorse the perspective of an "external" observer on my own experiences? What if I feel unreal, non-present, walking through daily life like an automaton, or like in a dream?

Depersonalization (DP henceforth) is a condition characterized by distressing feelings of being a detached, neutral and disembodied onlooker of one's mental and bodily processes or even of reality itself ("derealization") (Sierra & Berrios 1997). This profound disruption in the sense of self affects equally (a) the sensory and bodily aspects of the self (detachment from one's body or body parts); (b) the experiential aspects (detachment from one's subjective feelings and emotions); and (c) the cognitive and narrative aspects (disconnection from one's personal stories, memories, thoughts and future plans), often described by DP experiencers as a lack of a narrative flow or a "plot" in one's life. (Ciaunica et al., 2020).

When these phenomena become debilitating and chronic as dominant symptoms, they are diagnosed as depersonalization-derealization disorder (DPD) in DSM5, grouped among the dissociative disorders (American Psychiatric Association, 2013). DPD has been characterized as involving four main experiential components: (1) feelings of disembodiment (lack of body ownership and/or agency feelings), (2) emotional numbing (altered ability to attribute to perceived objects or concrete situations with emotional feeling), (3) anomalous subjective recall ([a] the "factual" aspect of memory is preserved but without the feeling the event really happened and [b] disturbances in the capacity for imagery), and (4) derealization (detachment, feeling cutoff from the world, things seem unreal; Sierra & David, 2011).

Transient DP experiences are reported in several major conditions such as (a) anxiety (Baker et al., 2003), (b) panic attacks (Madeira et al.,), (c) post-traumatic stress disorder (Lanius et al., 2012; Serra et al., 2007), (d) burnout syndrome (Mayer-Gross, 1935), (e) after prolonged sleep (Bliss, 1959), (f) hallucinogenic toxicosis (Fleming, 1936), (g) severe melancholic episodes (Fuchs, 2005; Kraus, 2002), (h) schizophrenic passivity experiences (Langfeldt, 1960); and (i) borderline personality disorder (Sar et al., 2017; Schmidt, 2021). In this paper however, we are concerned only with depersonalization experiences in the general population.

DPD has a prevalence of around 1–2% of the population (Hunter et al. 2004; Lee et al. 2012), which is comparable to that of schizophrenia for example. Also, the prevalence of transient episodes of DP is between 34 and 70% in the general population (Hunter et al., 2017). Transient occurrences

of depersonalization episodes have been reported by almost 50% of college students (Dixon, 1963). Everyday phenomena such as fatigue (Tibubos et al., 2018), sleep deprivation (Van Heugten–van der Kloet et al., 2015), or traveling to new places may also trigger transient depersonalization feelings (Kaplan et al., 1998; see, Salami et al., 2020 for a recent discussion).

In the light of this significant prevalence in the general population, it is important to delineate the phenomenological markers that characterize DP experiences. This is because when people start to experience these phenomena, they typically report them to close and trusted people. In other words, they use self-reports. There is thus an urgent need to better examine and understand DP experiences as reported from a first-person perspective. Qualitative studies have the potential to provide precious insights into the key aspects of phenomena as subjectively lived by the individual in day-to-day life. A more detailed understanding of the qualitative features of self- and world-experiences among those who experience high levels of DP can help to identify future targets for researchers looking to better understand the nature of DP in the general population.

The current paper proposes an exploratory qualitative study of anomalous self and world-experiences that occur in individuals with high DP experiences. It aims at providing an in-depth examination of self-reported disturbances of one's relatedness to one's self and the world, thereby shedding further light on the nature of altered subjective experiences among those with significant DP experiences.

In doing so, we build upon two influential qualitative research tools, namely (1) the Examination of Anomalous Self-Experience (EASE; Parnas et al., 2005) and (2) Examination of Anomalous Self-Experience (EAWE; Sass et al., 2017). The EASE and EAWE are two semi-structured phenomenological interviews derived from self-reports, clinical descriptions and interviews as well as empirical and theoretical studies designed to tackle disturbances in the lived self and world (though not necessarily exclusive to) schizophrenia spectrum conditions.[1] The EASE emphasizes disturbances in *self*-experience, whilst the EAWE is intended to investigate disturbances of the *world*-experiences. Phenomenological interviews like the EASE and the EAWE have been demonstrated to be highly accurate at identifying schizophrenia spectrum conditions and distinguishing them from other psychiatric disorders (Raballo et al., 2021). In previous studies using the EASE and EAWE with depersonalization/derealization disorder, the interview structures have captured a wide range of experiential changes in DPDR, and have demonstrated intriguing parallels and distinctions from schizophrenia spectrum disorders (Sass et al., 2013; Pienkos & Sass, under review). They are also effective at eliciting rich, nuanced descriptions of experience, often necessary for accurate clinical diagnosis and conceptualization, which may not be found in response to standard self-report measures or structured clinical interviews (e.g., Nordgaard et al., 2013; Sholokhova et al., 2022).

The idea that our everyday experiences are characterized by a basic sense of self, referred to as the "minimal self" has been highlighted by a longstanding phenomenological tradition (E. Husserl, J.-P. Sartre, M. Merleau-Ponty) and more recent authors (Gallagher, 2000; Metzinger, 2003; Zahavi, 2005; Hohwy, 2007; Blanke & Metzinger, 2009; Blanke 2012).

While the EASE/EAWE have the advantage of providing a comprehensive assessment of minimal self-disturbances, they may be also resource-intensive instruments, requiring significant training and potentially lengthy interview times. Hence, in some research situations, self-reports in shorter versions might be considered as an alternative option.

For example, the Inventory of Psychotic-Like Anomalous Self-Experiences (IPASE; Cicero et al., 2017) was developed within the tradition of objective scale development (Clark & Watson, 1995) in which an over-inclusive item-pool was established and items were retained or excluded based on their performance on exploratory and confirmatory factor analyses. Preliminary data indicates that the IPASE demonstrates strong convergent and discriminant validity, and that it may be used as an effective screener for disturbances of minimal selfhood (Nelson et al., 2018).

Recently, a new tool has been proposed, namely the Examination of Anomalous World Experience-Self Report (EAWE-SR). The EAWE-SR is a qualitative self-report tool developed to specifically reflect the content of the EAWE, with minor wording and organization changes to improve clarity and comprehensibility by participants without a background knowledge of phenomenology. Its psychometric properties have not yet been established at the time of the writing of this paper.[2] In the current study, its primary purpose (along with the IPASE) is to generate descriptions of a wide range of anomalous experiences. Therefore, its use is identical to other open-ended questionnaires, which do not typically undergo validation prior to use in qualitative research.

Taken together, these self-report tools offer an extremely comprehensive assessment of qualitative alterations in the experience of self and world. For this reason, they were adopted for this study as a means of eliciting descriptions of anomalous self- and world experiences among persons with high DP traits in a self-report format. Our study offers precious new insights into the self and world of those who experience high levels of depersonalization from a first-person perspective. These indicators may thus significantly advance our understanding of the altered sense of self and sense of realness among those with the capacity for high levels of DP. They can also be used in exploring further depersonalization, a widely spread yet under-studied condition.

Our paper proceeds as follows. In Sections 2 and 3 we describe in detail the Methods that we used in our study, the qualitative themes, as well as our Results. Section 4 provides a general discussion of our findings. Finally, in Section 5 we briefly present the limitations, conclusions and future directions for further research in this important and neglected area.

2 Methods

2.1 Sample

A total number of 24 participants (18 females and 6 males) participated in this study. Participants were recruited from previous studies conducted by members of this research group. Participants were invited to participate if they had high trait DP as determined by scores of higher than 50 on the Cambridge Depersonalization Scale (CDS-29; Sierra & Berrios, 2000) at the time of previous testing, and if they had agreed to be contacted for additional studies. The average age of the participants was 23.33 years (SD = 4.14 years) and participants came from a range of ethnic backgrounds (11 White, 4 East Asian, 1 Black and 8 South, West or Central Asian).

When recruiting for the initial study participants were recruited using the research institute database, university subject pool, and university participant mailing channels. Initial screening for DP was completed using an online version of the CDS-29 to measure the frequency and duration of depersonalization symptoms within the last 6 months. The CDS-29 contains statements describing abnormal sensory experiences and somatosensory distortions, inability to experience some emotions, heightened self-observation, autoscopy and out-of- body experiences, abnormal body ownership and lack of agency (Sierra et al., 2005). The frequency of these experiences has been measured on a 5-point Likert scale (never 0, rarely 1, often 2, very often 3, all the time 4), and their duration on a 6-point scale (few seconds 1, few minutes 2, few hours 3, about a day 4, more than a day 5, more than a week 6). All scores were added to compute each participant's global depersonalization score (0–290). Participants with a score higher than 50 were assigned to the High DP group and were invited to take part in this study. The CDS also provides the following subscores: Anomalous Body Experience (ABE), Anomalous Subjective Recall (ASR), Emotional Numbing (EN) and Alienation from Surroundings (AfS). These were used to assess the relation between CDS scores and IPASE scores.

We also included four additional screening questions: (1) "Do you have nightmares more than 4 times/week?"; (2) "Are you taking antidepressants or antipsychotics?", (3) "Do you see colours when you listen to music?", and " (4) Are you suffering from seizure disorder?"). An answer of yes to any of these questions led to exclusion on the study. Additional eligibility criteria included: (a) fluent in English; (b) between the ages of 18 and 40; (c) no history of neurological illness. These criteria while not entirely removing the possibility that our participants had mental health comorbidities did greatly reduce that possibility.

All participants were given a unique code and provided written informed consent prior to any data collection. Participants then completed all study measures online (using their own devices to complete the measures). Participants were notified before completing the measures that they were

allowed to stop at any time, that they could take breaks as needed, and that they could withdraw from the study at any point. Participants were also given the information to contact a crisis center or hotline prior to and after completing study measures as well as a link to information on depersonalization. All information was pseudo-anonymized by study personnel to protect the identity of participants. The study was given ethical approval by UCL's Research Ethics Committee.

2.2 Study measures

Study measures included the Inventory of Psychotic-like Anomalous Self-Experiences (IPASE) and the Examination of Anomalous World Experience-Self Report version (EAWE-SR). The IPASE (Cicero et al., 2017) is a 57-item self-report scale that was designed to quickly assess a subject's level of anomalous self-experience. It was developed by generating an overinclusive list of questions thought to capture disturbances of selfhood, which was then refined based on responses from large subject pools. Exploratory factor analysis has revealed five unique factors reflecting disturbances in Cognition, Consciousness, Self-Awareness and Presence, Somatization, and Transitivism/Demarcation. Each item includes five possible responses, which are scored from 1–5: *Strongly disagree* (1 point), *Disagree* (2 points), *Neither agree nor disagree* (3 points), *Agree* (4 points), and *Strongly agree* (5 points). Total scores on the IPASE may range from 57 to 285. Past studies have demonstrated strong psychometric properties, including correlation with high levels of positive schizotypy and schizophrenia, groups that are thought to also have high levels of self-disturbance (Cicero et al., 2017). For the purposes of retrieving qualitative information, an additional open-ended response was added to each item: *What is the experience like for you? Please provide any details you think are relevant to understanding this experience.*

The EAWE-SR (unpublished) is a 210-item self-report instrument that was developed on the basis of the EAWE interview (Sass et al., 2017). It assesses six domains of anomalous world experience: Space and objects, Time and events, Other persons, Language, Atmosphere and Existential orientation. Each item includes four possible responses, which are scored from 1 to 4: *I have never experienced this* (1 point), *I have only experienced this occasionally* (2 points), *I have experienced this frequently* (3 points), and *I experience this every day or constantly* (4 points). Scores may range from 210 to 840. In this study, all items also contained the following open-ended response option: *What is the experience like for you? Please provide any details you think are relevant to understanding this experience*. As this instrument is still in development, no psychometric data is available; it was used solely for the purpose of eliciting qualitative data about a large range of potential changes in experiences of the external world.

2.3 Data analysis

Basic descriptive statistics (mean, standard deviation, range) for both the IPASE and the EAWE-SR were calculated. As the psychometric properties for the EAWE-SR have not yet been established, descriptive data are presented for informational purposes only.

Qualitative data were analyzed using inductive thematic analysis (Braun & Clarke, 2006). This approach to data analysis is used to organize similar pieces of data around different themes that emerge from the data itself. In this study, it involved several steps: familiarizing oneself with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report. In this study, two coauthors (EN and LM) reviewed portions of the data and provided reflections on data meaning. Following this, one coauthor (EP) developed codes based on the data and reflections and used these to generate themes. Responses were not coded when they were clearly describing experiences that only occurred in the context of substance use, a medical condition, sleeping/waking up, or as the result of imagination or playful entertainment of ideas. They were also excluded if they reflected very "typical" experiences according to at least two authors who reviewed the data – that is, experiences that occur as part of common, everyday experience, though unusual experiences that may be encountered by people outside of pathological conditions were coded (e.g., feeling like one has a special purpose in life: "Sometimes I think things that are happening to me are related in some way with my purpose in life," P236; or having a strong preference for order and symmetry: "I like things to be symmetrical and the objects that surround me to be constant," P149). Note that experiences that occurred in the context of altered emotional states (e.g., depression, anxiety, euphoria) were included, though the description of the results note when these emotional states were found in participant descriptions. All themes were then reviewed by EP, with changes made as necessary to ensure coherent and cohesive themes. Themes were then defined, and themes and coding scheme were sent to coauthors (AC, EN, and LM) to review and provide feedback on appropriateness of code and theme identification, and suggest changes as necessary. These suggestions were reviewed and any discrepancies or disagreements were discussed by coders to arrive at consensus.

3 Results

3.1 Descriptive statistics

The means of total scores on study measures are reported in Table 1. Some CDS scores were lower than those reported during the initial screen for the study – and indeed, some scores fell below the initial cutoff of 50 points. However, other scores were higher, suggesting that any changes in scores

Table 1. Means and range of study measures.

Measure	Total score (SD)	Range
CDS (start of study)	63.75 (25.84)	17–115
CDS (pre-study screen)	73 (25.01)	51-151
IPASE (total score)	123.33 (31.79)	71–180
IPASE (item score)	2.16 (0.56)	
EAWE-SR (total score)	291.79 (52.56)	215–410
EAWE-SR (item score)	1.39 (0.25)	

represented natural fluctuations of DP experiences over time. All data was kept for this study, as the primary goal was to include participants who had experienced high levels of DP at some point in their lives (though not necessarily at the time of data collection). All except one participant indicated that they endorsed one item on the CDS as occurring "often," "very often," or "all the time," indicating that some experience of depersonalization continued to occur regularly in their lives.

3.2 Qualitative themes

The following themes reflect a diverse range of experiences reported by the participants in this study in response to the open-ended questions added to the IPASE and the EAWE-SR. Each theme comprises a number of related codes, which are also listed. Detailed descriptions of the codes and illustrative examples are provided in the supplementary material, with a selection of these included in the text below. The number of participants who describe such changes are listed in parentheses, and illustrative quotes are identified by the participant number following the quotation.

3.2.1 Detachment (22 participants)

Many participants described feelings of being detached from the world and from oneself, involving a sense of not being fully engaged with or thrown into the world and one's own experience. The clearest examples of detachment involved a general sense of not being fully present to oneself, of feeling like an observer of oneself or the world, of not feeling a sense of ownership over one's own thoughts and experiences, or of being detached from or not a part of the world ("I just feel like i'm watching myself over a distance and I don't know who I am," P218). Other experiences included in this theme were feelings of objects or places being unfamiliar, or simply marveling at their existence, and of the world feeling unreal or lacking in vibrancy or depth ("Sometimes it feels like I've been picked up out of my real world and put down into a film set, or something – as though it's a copy of the real world but not quite real," P241). It also includes feelings of *déjà vu*, as though reliving a moment rather than living fully in the moment. (It is worth noting that some participants reporting *déjà vu* appeared to have experiences similar to the general population; though for some, these feelings could be quite strong and confusing.) Some

participants noted that these feelings appeared to occur primarily in situations in which they were feeling depressed or anxious, though others did not attribute their experiences to particular psychological states.

3.2.2 Embodiment (14 participants)

There were many descriptions of unusual or changing experiences of the body. These could involve a fairly profound feeling that the body is disconnected from or not belonging to one's self, feeling as though one's expression or the body is out of one's control (e.g., feeling unable to move the body intentionally; "whenever I feel intimidated, I tend to lose control of my body and

Can't move properly," P139, or the body moving without apparent intention to do so; "Sometimes when I was focusing on a specific task such as typing out documents, my hands do not feel like my own," P215), that it has changed or is in the process of changing inexplicably ("sometimes it feels as thought i've shrunk down and everything seems to big, and the room seems so big," P102), that the participant is experiencing physical sensations that do not seem to have a clear physical cause ("Sometimes it feels electric, as though there's a current running though me," P241), or sudden and inexplicable feelings of bodily weakness.

3.2.3 Identity changes (19 participants)

Participants described a lack of congruency or transparency related to one's sense of self. This could include a feeling of distance from oneself ("I sometimes feel separate from my own existence," P124), an uncertainty about who one really is, a lack of congruency between one's felt self and one's thoughts or words or actions ("My thinking self and my speaking self feel different," P217), or a feeling of disconnection or lack of continuity with one's past self. It also included feeling like one did not identify with one's age, feeling either older or younger than one actually is ("I feel older than I am," P149).

3.2.4 Idiosyncratic beliefs (20 participants)

Participants described holding several unique preferences, attitudes, beliefs, and values regarding their relationships, society, humankind and, most broadly, existence and reality itself. They described feelings of indifference toward others or humankind or the world as a whole ("I often feel like I cannot empathise with the tragic things that happen in the world and I have to pretend to care about these things," P158; though note that some described a feeling of greater concern or responsibility for others, described below), a tendency to be skeptical of accepted aspects of reality ("I doubt of everything. I doubt about the real feelings of my friends and my boyfriend towards me, sometimes I even doubt about my own and other's existence," P236), and a rejection of conventional attitudes or behaviors (either due to a preference for one's own values or dislike of conventional attitudes and

behaviors), while others noted a preference for things and events to be orderly and regular. Some described a sense of having greater insight or awareness than most, a feeling of being guided or predestined in life, and a sense of encountering events that conveyed a special meaning to them. Several participants described anomalous ways of thinking, such as having greater openness to seeing similarities or connections that others might not recognize, looking for the reason or explanation for every event, and forming beliefs without apparently sufficient supporting evidence. A number of participants also described beliefs about the end of the world being imminent ("I feel like the end of the world could happen any time and it still give me panic attacks," P248), while some discussed a general awareness of a larger perspective than just their immediate view or that of the human race. About half of the descriptions (53 out of 103 codes) were characterized as something that was *typical* of the individual (e.g., "I tend to . . . " "I like to . . . " "I prefer . . . "), while the other half (50 out of 103 codes) were described in particular contexts or in response to specific events (or using terms like "Sometimes," or "Once in a while"). In both types, many of these ideas, beliefs, and values appeared to be a *tendency* of the individual (rather than, e.g., a state-based phenomenon only associated with anxiety or DP) although one that may only be triggered in specific contexts.

3.2.5 Affective changes (21 participants)

Participants described various forms of emotion and mood that were different from what the situation would typically elicit, including anxiety (which many described as occurring for no apparent reason; "I generally live very anxiously, even without reason, and even when I could live more relaxedly, I always manage to find something that causes me anxiety," P201), restlessness, and rapid shifts in emotions or emotions that are felt to be "inappropriate" to the situation ("I can be happy and then agitated a few minutes later for no real reason," P216). Related to this were specific experiences characterized by a sense of demoralization, including hopelessness ("Feels like it's not worth trying and just let life takes me whenever it wants," P230), feelings of being unable to achieve one's goals or direct one's life, feeling like life is meaningless, overbearing feelings of fatigue, and not feeling strong emotional reactions or excitement about anything ("I feel as though I have lost some part of myself that used to get excited at the smallest things," P153). One participant also described concerns about losing their sanity.

3.2.6 Changes in communication and approach to language (18 participants)

Participants noted numerous difficulties in expressing oneself clearly or fully to other people, or else the use of certain strategies to enhance clarity. This includes problems with finding the right word or clearly articulating oneself, lacking emotional intonation in speech, feeling unable to organize one's

thoughts effectively in speech ("There are so many things I am thinking about that I don't know how to verbalize them so that they make sense," P147), or no longer wishing to speak. Some participants also described shifts in the way they think about, understand, or approach language itself. This includes a feeling of words being unable to convey experiences ("Some of my experiences are very difficult and words cant properly describe them," P136), various difficulties understanding speech or written language ("sometimes I just cannot distinguish what was being said, it just sounded like gibberish, P230), feeling distracted by certain words, feeling like language and words are meaningless or absurd, or feeling like language has a certain degree of power or autonomy.

3.2.7 Anomalous reactions to others (21 participants)

Participants described several habitual and automatic reactions to other people. These might occur as a stylistic preference, a personality trait, or in response to reactions that others elicit. These experiences include rejecting some social norms (or preferring one's own norms), acting clownishly, acting in oppositional ways ("I can be stubborn and will resist for no personal reason," P154), and habitually avoiding the company of others (often due to feeling overwhelmed or exhausted; "I would avoid any interaction because it is easier for me to do that. Have conversations takes a lot of energy from me and it's easier to avoid them," P136). Others described feeling strongly influenced by others, at times intentionally imitating others' behaviors and mannerisms, or else feeling like their entire identity was shaped by other people ("I think I am pushing off my identity and just taking on things I see other people do or say," P221). Some described acting in ways that were counter to their own preferences in order to be liked and to fit in or else in order to avoid conflict.

3.2.8 Difficulty relating to others (21 participants)

Participants described feelings of being disconnected or distant from other people, including feeling profoundly alone or isolated, either due to general feelings of distance and detachment ("Even though i'm seated with a group of people, I feel like i'm not even there or there's a wall that separates me from everybody else," P218), not understanding others ("I simply cannot understand people, anymore," P203), feeling the need to scrutinize and observe others' behavior to understand them, having to apply intellectualized strategies to understand and interact with others, not feeling truly known by others ("I feel like they don't know me and I don't know anything about them" P248), feeling that others put on a mask or facade, or generally disliking other people. Some described feeling like people who were known to them were suddenly unfamiliar (to the extent, for one person, of doubting their past experiences with that person), feeling like others are lifeless or

unreal ("A few times I've had the experience of being in a crowd of people, and feeling as though some of them were zombies or clones or mannequins … I have to watch them for a little while to make sure they do something human-like," P241), or feeling like some people are especially and unusually lively.

3.2.9 Social anxiety and self-scrutiny (21 participants)

Participants described numerous experiences of anxiety, fear, self-doubt, and self-criticism in social situations. These could occur during specific interactions with individuals, but they could also characterize an approach to other people as a whole. These experiences included feeling unable to respond adequately in social situations ("I feel like I have no social skills and have no idea how to act upon them," P243), feeling uncomfortable and pressured in one's interactions with others, feeling uncomfortable specifically related to making eye contact with others ("I sometimes find eye contacts very unsettling, like vulnerable feeling or intrusive feeling," P230), generally feeling self-critical and inferior to others, feeling like others are especially focused on or watching/observing one ("I always feel like my entire life is being watched by observers, judging my behaviour," P217), experiencing others as judging one negatively, intentionally hiding one's reactions in order to avoid exposure, and mistrusting or viewing others as threatening or intending one harm in some way. Related to this was a tendency to scrutinize oneself and feel self-conscious.

3.2.10 Unusual experiences of time (22 participants)

Participants described numerous changes in their experiences of time, including both what might be characterized as implicit time (how time is experienced in the moment) as well as explicit time (how we reflect on our past or likely future; Fuchs & Van Duppen, 2017). This includes a variety of changes in immediate perception of passage of time and of the speed of one's own movements or the world, such as feeling like one moves more quickly or more slowly than others, or like time itself is flowing faster or slower than it seems like it should ("Sometimes I feel like time is moving incredibly fast and I lose the perception of time," P136), or feeling like time is leaving one behind. Some described trouble organizing events that are happening in the present or in the past, a feeling that time has stopped or disappeared ("Watching TV I feel like I go numb . . . I go into a black hole of time somehow," P221), as well as feeling like one is only to focus on or think about the present moment (and so unable to reflect on or integrate the past or the future). Some described feeling disconnected from the future ("What I do in life right now just doesn't seem to be leading up to the future," P218): like the future might not occur, was irrelevant, or was impossible to affect with one's actions. Others found the future to be threatening and anxiety-

provoking, or generally had a sense of being able to predict or anticipate future events. Some participants described a sense of past events feeling either stretched out and lasting longer than they actually did, or else having moved very quickly or being much more recent than reality.

3.2.11 Changes in perception and sensation (23 participants)

Participants described many different changes to sensation and perception. Many noted changes to the intensity of their sensations: sounds, tastes, textures, smells, colors, and general brightness could all feel more intense or less intense than usual. One's vision could be blurred ("like someone has put dirty fingerprints on my sunglasses – but its my eyes," P206) or one could feel temporarily or partially blinded, sounds could seem distorted (especially human voices), or one may be especially sensitive to background noises. The qualities of sensory objects could also change: shapes may look different, the size of objects may appear larger or smaller than reality, apparent colors may change, and some textures may feel different (such as feeling dirty or less solid than reality). Some described these changes as seeming to emanate from the sensory objects themselves (e.g., music sounded louder: "Sounds do seem to be louder for me than others I am with . . . For example at the gym once I asked them to turn down the music but they said it wasn't loud, but I couldn't stand it," P102), but many described the changes as related to their own sensory apparatus. Such changes tended to reflect a *heightened* sensitivity to various senses (especially vision, audition, and taste), though some noticed a tendency not to notice certain things (see also cognitive organization below). A number of participants also described changes to their ability to accurately or effectively perceive objects, including difficulty identifying the shape or sound that one is perceiving, difficulty identifying the direction or location of the origin of a sound ("Voices can also sound like its coming from further away at a distance even though the person is talking directly to me," P218); also including feeling as though sounds were not coming from the object that produced them), and difficulty with the perceptual integrity: either feeling unable to perceive objects or landscapes as integrated wholes (rather than a set of isolated elements), or else difficulty seeing distinctions between separate objects. Specific changes to the perception of space were also noted, including problems with the accurate perception of distances (objects seeming further away or closer than they really were; "When I look at something from a mid-long distance, sometimes it seems as though it was further away, as though it was out of reach even when I could pick it up without moving," P243), feelings of disorientation (including getting lost in familiar places), a feeling of space expanding or else closing in, a feeling of objects or space being flattened, an increased awareness of space itself, and a sense of spaces and places having moods emanating from them.

3.2.12 Confusion of the boundaries between self and world (22 participants)

Participants described various feelings related to a blurring of the borders between self and non-self (either other people or, occasionally, objects). Some experienced a kind of privileged access to others' thoughts and private feelings, while other participants described the feeling that other people had control over their private thoughts or feelings ("I feel like someone is telling me what to do and say," P216), else that their thoughts or feelings were somehow public and exposed to others, or that close contact with others could infect or contaminate them in some way. A small number of participants described feeling like objects had merged with their body somehow, like objects or the world were communicating with them in some way, or that external sounds were coming from inside their head (or mind). Similarly, participants also described vivid experiences (similar or identical to perception) of things that are not actually/currently present. Some described this as having especially vivid imagination or visualization, others as having a very intense experience of past memories ("sometimes I get confused as if the present was re-living a past memory," P226), some generally feeling confused about whether they were imagining or thinking something vs. perceiving or remembering it ("When i'm having a phone call with someone, it's hard to imagine if there's anyone on the other side of the line or if i'm just imagining a voice talking to me," P218). Some noted that they could perceive sounds or images that were not actually present – either continuing to hear or see something that was previously, but no longer, present, or perceiving any number of precepts that were not present (including vague shapes, specific objects, geometric patterns, spots or dots, orbs of light, noises, voices, or smells).

3.2.13 Disturbances of attention and cognitive organization (21 participants)

Participants described several difficulties related to the ability to effectively attend to, think about, or remember things in order to achieve desired goals. This could include attentional difficulties, such as not noticing things that were obvious to others or else becoming distracted by irrelevant details or one's own thoughts ("Very often, I only hear half of what is said to me. I try very hard to bring myself in the here and now. My mind wanders away without my control really," P220), as well as being highly aware of repetitions or patterns of events or objects (for no apparent purpose), feeling as though one has too many thoughts or that thoughts are moving too quickly to be able to follow them effectively, feeling unable to effectively make decisions because of taking too many factors into consideration ("I have a lot of trouble making decisions and in situations, I try to see at least two different sides of the situation," P158), and having difficulty remembering past events ("Things that actually happened yesterday or the day before that I cannot remember.

I cannot remember what I did or where I were," P236) or remembering to do something in the future, which was often related to feeling present or attentive to the original experience or thought.

3.2.14 Greater existential concerns about others and the world (17 participants)

Some participants described a sense of stronger concern toward others and the world, or an experience of positive emotions associated with experiences. This could include a general sense of heightened connectedness to the humanity or the world ("I was a part of everything, and the whole world, sometimes the whole universe was in a sense a part of me," P147), as well as feeling especially close to or familiar with people who the participant was in fact unfamiliar with, a stronger sense of empathy toward and understanding of others, a heightened sense of concern or responsibility for caring for other people or for the world/humanity as a whole ("I do have the responsibility to make a change and have an impact in the world," P236), and a sense of being aware of and appreciating the beauty or other qualities of one's environment. These findings are in line with previous work highlighting greater existential concerns in people with depersonalization experiences, including concerns about the meaning of one's presence in the world, enhanced sense of observations of others' behavior and character, greater concern for the purpose and path of the humanity as a whole.

3.3. Quantitative analysis

A quantitative analysis was also conducted on the total IPASE scores. We conducted a four-stage hierarchical multiple regression with the total IPASE score as the dependent variable and the four CDS subscores (ABE, ASR, EN and AfS) as predictors. The subscales were entered in an order that seemed plausible given their relationship to self-perception, following (Farmer et al., 2020). Full results of the hierarchical regression are given in supplementary materials (see, Table 2). Only the AfS significantly improved the model, explaining an additional 25.4% of the variation of the total IPASE scores compared to the model including ABE, ASR and EN as predictors. The AfS was found to be a significant negative predictor of IPASE total score ($ß = -0.755$, $p = 0.007$).

4 Discussion

Feelings of depersonalization (DP) are extremely common in the general population, yet under-acknowledged and under-examined. Given that DP seems to affect the subjectivity of one's experiential life, inducing profound

Table 2. Summary of the hierarchical regression analysis for CDS subscales predicting IPASE total scores.

Variable	ß	*p*	R^2	ΔR^2	ΔF	Δp
Step 1	0.334	0.111	0.111	0.111	2.760	0.111
Anomalous Body Experience						
Step 2	0.276	0.243	0.124	0.13	0.311	0.583
Anomalous Body Experience	0.128	0.583				
Anomalous Subjective Recall						
Step 3	0.065	0.804	0.222	0.098	2.515	0.128
Anomalous Body Experience	–0.062	0.809				
Anomalous Subjective Recall	0.463	0.128				
Emotional Numbing						
Step 4	0.326	0.179	0.476	0.254	9.214	0.007
Anomalous Body Experience	0.041	0.849				
Anomalous Subjective Recall	0.748	0.010				
Emotional Numbing	–0.755	0.007				
Alienation from Surroundings						

disruptions in self-awareness (Sierra & David, 2011; Mansell & Carey 2013; Deane et al. 2020; Gerrans 2020; Limanowski & Friston 2018; Ciaunica et al. 2020; Ciaunica et al. 2022), a phenomenological and qualitative approach may open a new window onto the very core of these experiences as lived from a first-person perspective.

In the current study we combined a well-established qualitative tool (IPASE) with an innovative tool (EAWE-SR) in order to provide key new insights into the experience of DP as lived from a first-person perspective. To the best of our knowledge, this is the first qualitative study examining anomalous self- *and* world-experiences in individuals with DP traits. As we will see shortly, exploring how DP is embedded in not only a surprising array of (a) *self*-perception disturbances – that is, the way people relate to their own self and body – but also (b) *world*-perception disturbances – that is, the way people relate to the world and others – offers a richer and more complete picture of the phenomenon. That is, by having a clearer picture of the nature of self and world among those who experience high levels of DP, we see that DP may occur within a particular subjective context, one which is not typically explored beyond the acute features of the DP episode. The exploration of these experiences presents numerous potential targets for future research on transient DP.

The descriptions from our participants with high levels of DP traits indicate the presence of a wide range of anomalous experiences, including those that are traditionally understood to be core features of depersonalization, as well as experiences that are not typically included in the depersonalization construct. Indeed, our findings may be clustered in two main groups: (a) anomalous self- and world-experiences usually associated to depersonalization traits and widely reported in previous research; (b) new experiential features that are less highlighted in previous work, and which

may offer essential insights into the experience of DP, thus drawing attention to key aspects that otherwise may be overlooked in understanding depersonalization.

In what follows we focus on the qualitative analysis and describe the main experiential clusters and the reported disturbances in turn, illustrating how our findings provide key new insights into the subjectivity of those who experience depersonalization from a first-person perspective.

4.1 Anomalous bodily experiences and emotional flatness

In a comprehensive review of historical cases of pathological depersonalization, Sierra and Berrios (2001) found five major domains of experience that were consistent over time: changes in body experience, loss of agency, emotional numbing, changes in mental imagery and memory, and changes in visual perception of surroundings. Other studies (Sierra et al., 2005; Simeon et al., 2008) have found that these experiences separate into several distinct factors, including anomalous body experience, emotional numbing, anomalous subjective recall, and alienation from surroundings, with one study finding a fifth factor termed "body distortion" (Simeon et al., 2008). The results of the current study map onto these features and factors, particularly the changes in embodiment (which also includes disturbances of agency), detachment, affective changes, changes in perception and sensation, and cognitive organization (specifically related to disturbances of memory).

Anomalous bodily experiences have been consistently reported in relation to clinical and non-clinical depersonalization both in theoretical and experimental previous work (see, Salami et al., 2020 for a recent discussion). Neuropsychological tests have noticed wide perceptual and attentional alterations in the pathophysiology of depersonalization-derealization (Guralnik et al., 2007, 2000). Impaired bodily self-awareness has been linked with disrupted physiological response in patients with DPD compared to healthy participants (Dewe et al., 2018; Owens et al., 2015; Sierra et al., 2012). DP has also been linked to disrupted activity in neural regions underlying somatic processing (Lemche et al., 2013; Lemche, Surguladze, et al., 2016; Medford et al., 2016). Recent studies found atypical integration of self-related visuotactile information (Adler et al., 2016; Farmer et al., 2020; O'Sullivan et al., 2018). Atypical neural activity have been reported in DPD patients with regards to sensory information (Medford et al., 2016) as well as regions responsible for the processing of visceral signals (Lemche et al., 2013). Interoception is typically defined as the perception and processing of visceral bodily signals, and the main brain regions responsible for processing interoceptive inputs are insula, cingulate cortex, amygdala and somatosensory cortex (Craig, 2009). Crucially, all these regions have been linked with atypical activity on DP and DPD patients (Sierra et al. 2014;

Medford, 2012; Medford et al. 2016). This led some theorists to examine links between disembodiment and interoceptive processing in DP (Michal et al. 2014; Seth et al. 2011; Gerrans2018; Ciaunica et al., 2021a, in press, b).

In our study, participants consistently reported feelings of "disembodiment", and consequent emotional flatness and "emptiness". For example: "It feels like an out of body experience" (P216) or: "I feel like I'm watching from inside a shell. Everything besides my limbs feel like they don't have much of a presence" (P 217). Even more strikingly: "I feel nothing. I just observe" (P 243) or: "People call me poker face. I often have trouble to show that I am happy from my tone or expression" (P215).

Although data from this study cannot demonstrate causality, previous research suggests that these feelings of disembodiment, and of being an "onlooker" or "watcher" or "observer" of one's experiences are also impacting one's sense of agency: "I have a feeling of not actually doing something even though I am about to do it. I know what I am supposed to feel, but it does not come from me organically" (P 221). Lack of immersiveness or presence in the world makes people feel unreal, and seem to act like automata: "I sometimes feel like I'm not really present and I feel mechanical" (P124). Feelings of inner "emptiness" are frequently reported: "I tend to feel as though it is empty inside and get tired from doing nothing" (P 153) or: "There's a physical feeling of emptiness in my chest" (P218) and: "I feel empty inside" (P248).

Previous work has proposed a link between the attenuation of the emotional response in DP and (a) *decreased* activities in emotional cortical (insula) and limbic (hypothalamus, amygdala), while (b) *increased* activities of the prefrontal cortex (Lemche et al., 2007; Medford et al., 2016; Sierra & David, 2011). Our findings again support the idea of a strong connection between anomalous bodily experiences and emotional numbing in DP. These experiences seem to be underpinned at the neural level by a decrease activity in limbic areas (consistent with the disembodiment and emotional flatness hypotheses) and an increase in prefrontal activity (consistent with the "hyperreflexivity" hypothesis). We turn to this discussion now.

4.2 Disturbances of the minimal self and hyperreflexivity

Several theorists draw on a long-standing phenomenological tradition to define the minimal self as defined as the pre-reflective and immediate experience of one's body, actions, perceptions, and thoughts in relation to the world (Ciaunica, 2016; Ciaunica & Fotopoulou, 2017; Fuchs, 2005; Legrand, 2006; Parnas & Handest, 2003; Zahavi, 2005). For example, L. A. Sass and Parnas (2003) have hypothesized that a *basic self-disturbance* (BSD) (also termed self-disorder, ipseity disturbance, or disturbance in the "minimal self") may lie at the core of severe psychiatric conditions such as schizophrenia and

schizophrenia-spectrum disorders. L. A. Sass and Parnas (2003) have identified at two core features of BSD, namely: (a) "*diminished self-affection or self-presence*" which refers to a diminished sense of being immediately present to one's experiences; and (b) "*hyperreflexivity*" which refers to forms of heightened self-consciousness and heightened awareness of typically tacit or implicit aspects of one's experiences.

Much emphasis has been placed on diminished self-presence and altered self-awareness in DP (Sierra & David, 2011). However, it also seems that hyperreflexivity may play a role in developing and maintaining DP symptoms. This over-scrutinization triggers simultaneously inner turmoil and non-responsiveness to the external world. Hence, diminished sense of self and presence and hyperreflexivity go hand in hand, like the two faces of the same coin (Ciaunica et al., (2021b, under review); Fuchs, 2005). Our study provides further support for this hypothesis. Indeed, our participants frequently reported these types of hyperreflexivity and existential turmoil. For example: "I sometimes feel separate from my own existence" (P124) or: "I'm not sure what the nature of existence might be, because all of the tools we would use to question it are part of this existence, so would naturally prove that we do, but how would we know if we were just the dream of a sleeping god or something?" (P241) and equally: "I feel that the universe has a special meaning and if I discover this meaning I will stop to exist. Sometimes I think people do not exist and even doubt about my existence at all, like if I were in a dream." (P236) or: Sometimes I'm so preoccupied with the way I walk or the way I present myself, I don't realize what's happening around me, for example if someone appeared next to me" (P218). These observations are in line with a wide body of research on DPD noting that individuals tend to focus on inner experiences and thoughts rather than attending and engaging with the external world (Hunter et al. 2003). Obsessional self-check of one's internal states: "how do I feel now?", "'Who am I"? DPD patients are drawn to ruminative and over-intellectualization of their inner workings (Hunter et al. 2003; Medford et al., 2005). As seen in the current study, it appears that individuals prone to high levels of DP are similarly prone to such intensive self-reflection.

This hypothesis is further supported by the results of our quantitative analysis, suggesting that individuals with higher self alteration (as measured by the IPASE) show less alienation from surroundings (AfS, measured by the CDS). Speculatively, one may argue that this negative correlation underwrites a tendency in people experiencing a greater degree of self disturbance to be less aware of their surroundings, thus less likely to notice (or potentially experience) acute feelings of alienation from the world.

Previous phenomenological work has suggested that atypical self- and world-awareness in DP alters the *transparency* of basic, pre-reflective forms of self-awareness, which triggers consequent feelings of self-opacity and unrealness or lack of presence (Ciaunica et al., 2020, 2021a, in press;

Fuchs, 2005). Transparency is an interesting and peculiar property of our experiences which has been theoretically spelled out in different ways by different theorists (Moore, 1903; Harman 1997; Tye, 1999; Metzinger, 2003; Fuchs, 2005; Ciaunica et al., 2020) and a detailed review of these accounts[3] lies beyond the scope of this paper. In a nutshell, transparency can be intuitively grasped via the so-called "window" metaphor. For example: a perfectly clear and transparent window glass or sliding door can give us the illusion of an unmediated access to a landscape, say. The landscape seems present and reachable paradoxically because the window's glass is transparent, invisible and taken for granted: it is there without us being aware that it is there. However, in some cases, as we will see later, one *can* become aware of the existence of the invisible and mediating transparency of the window itself so to speak. Indeed, suppose there is a crack in the pane of glass. Two important observations emerge here: a) first, we become aware of the cracked window itself as a *visible* observable entity: we realize that there was something there without us being aware of its presence in the first place. b) Second, while the cracks in the window's glass make the latter visible, they also make our access to the landscape more "opaque" or "mediated". We may still perceive the outer landscape through the cracked window, but its clarity is hindered. Now we are aware that something stands in the way, and disrupts our full immersion into the reality of the landscape. In sum, two key points are to be retained: (a) the property of transparency enables the subjective feeling that we are in immediate contact with our self and the world, and (b) that consequently both self and world are felt as being real and present.

The key idea embedded in this metaphor is that in DP, altered self-experiences or diminished self-presence – that is, the "crack" in the transparent and typically unnoticed window of one's subjectivity – leads to hyperreflexivity. This form of enhanced awareness of typically tacit or implicit aspects of one's experiences trigger alienating feelings of "opacity" are often and strikingly reported by DPD patients as "having a pane of glass" or "veil" interposed between one's self, body and the world, or "living in a bubble", or "feeling surrounded by an invisible wall" (Ciaunica & Charlton, 2018; Ratcliffe, 2017; Simeon & Abugel, 2006).

This over-scrutinization of one's inner workings may be accompanied by feelings of being "trapped" in one's head (mind) and outside one's body (Ciaunica et al., 2020; Fuchs, 2005). Interestingly, our participants in this study frequently described similar feelings. For example: "I am very in my own head and sometimes so focused on myself, that I miss the surroundings or miss what is going on around me" (P136) or: "Sometimes I do feel like my mind has a mind of its own and does what it wishes while my body is not connected to it"(P136).

Taken together, these findings support the idea that a profound disruption of self-awareness lies at the heart of the depersonalization experiences as reported by previous research (Sierra & David, 2011; Seth et al. 2012; Billon, 2016, 2017; Gerrans 2018). We also argue that depersonalization interferes with the *transparency* of one's minimal self, and that this is linked to feelings of unrealness, lack of presence, or distance between oneself and one's experiences. This lack of transparency was most obvious in the feelings of *detachment* described by most (though not all) participants, as well as in the feelings of disconnection from one's body and physical movements.

4.3 Insights into overlooked features of depersonalization experiences

Up to now we have examined our findings through the lens of previous work and hypotheses. We illustrated that the self-reports analyzed here support indeed the idea that anomalous bodily experiences, emotional flatness and consequent alterations of the minimal self and hyperreflexivity be considered as core features of DP experiences.

Interestingly, though, it is worth noting that there were many features reported by participants that are not described in the literature on depersonalization experiences, and that are not captured in common measures of depersonalization like the CDS. For example, various forms of "blurriness" or confusion between oneself and the world, such as being unsure about whether something was objectively perceived or simply imagined or thought, were described by numerous participants. One participant described the following: "When I'm having a phone call with someone, it's hard to imagine if there's anyone on the other side of the line or if I'm just imagining a voice talking to me" (P218). Other experiences in this category included feelings of having access to others' thoughts and feelings, or others having access to or being able to control one's own thoughts and feelings, as well as feeling "contaminated" by other people. Participants also experienced a kind of merging with objects in the world or feeling like objects were communicating with them. Furthermore, some experiences of other people could be so unusual as to call into question the realness of others: "A few times I've had the experience of being in a crowd of people, and feeling as though some of them were zombies or clones or mannequins, or something like that. It's an eerie feeling and I know that they are people, but I have to watch them for a little while to make sure they do something human-like. Sometimes it feels as though people's movements are programmed or studied versions of human movements" (P241).

Many participants also described disturbances in temporal organization, particularly with ordering events that occurred in the past. For example, "The last week in my memory is a few events with no connection, sometimes

I really can't imagine what else I did in the last day/week" (P221). Temporal perception seems here to be of particular interest, with many participants reporting that the time seems to fly faster than before experiencing DP, e.g., "Sometimes I feel like time is moving incredibly fast and I lose the perception of time" (P136).

We also found alterations in goal-directed thinking and attention (cognitive organization), described by one participant as, "Very often, I only hear half of what is said to me. I try very hard to bring myself in the here and now. My mind wanders away without my control really" (P220). Another stated, "Especially with my thoughts, they seem to jump about and miss steps out and I can get very lost within them" (P241).

Importantly, we noticed a heighted sensitivity to various sensations from different sensory modalities. One participant stated, "I find myself wincing from light colored things especially" (P221), while another noted, "The voice of those around me sometimes seems so loud that my ears pulsate" (P201).

It would be interesting and important to examine more closely in future work both the sensitivity to distinct sensory inputs (visual, auditory, tactile, olfactory, vestibular, proprioceptive) and their impact on time and space perception in people with DP experiences. Specifically, it would be key to investigate to what extent feelings of (un)realness are triggered by atypical sensorimotor integration of self- and world-related information. For example, it has been recently suggested that different senses and modes of perception (e.g., vision versus touch) may provide a different basis for self- and world-experiences (Ciaunica et al., 2021a, in press). Distant visuospatial perception – such as seeing/recognizing oneself in the mirror – may enhance feelings of estrangement and self-detachment, of being inside one's head. While active engagements with the world and others via proximal (tactile and multisensory) interactions may enhance the sense of self, realness and presence in people with DPD. This is because, paradoxically, in order to get closer to one's self, one needs to be able to "forget" oneself (Ciaunica et al., (2021b, under review)) and to be able to open to the world and others, via proximal tactile interactions.

As one person with lived experiences of DP strikingly puts the point: "When the depersonalisation is very deep, I still seek to 'be' someone else because it feels like that constant source of interaction is the only thing that allows me to maintain a connection with the world. I'll also seek physical contact with whoever I'm with. It almost feels as though I need to be that other person because my own sense of self is not strong enough in that moment to sustain me." (Ciaunica & Charlton, 2018). Indeed, this observation may be key for potential interventions and therapies designed to make DP people more aware of what still *connects* them with their self, body and the world, as opposed to focusing on what *disconnects* them from one's self

and others. Specifically, future work needs to disentangle and contrast the role of proximal (touch, interoception) versus distal (vision) senses in shaping the experience of being a self present in the world and fully connected with one's body.

Participants also described many disturbances related to self-concept and to relationships with others, including feelings of inferiority and insecurity, feeling uncomfortable around or exhausted by others, feeling disconnected from others, and adapting one's behavior to elicit others' approval; as well as more profound changes such as uncertainty about one's identity and lack of continuity in time. For example: "I can't connect the me right now to other 'me's throughout time" (P217).

Indeed, our study suggests that no domain of experience may go untouched: the passage of time, relationships with others, language and communication, attention and cognitive organization, and boundaries between self and non-self all become more blurred, more disorganized, less transparent, and less functional. Furthermore, thoughts and beliefs about oneself and others also seem to be deeply affected, as participants frequently felt insecure, self-critical, and uncomfortable, anxious, or mistrustful in social situations. Crucially, mood and affect were also altered beyond typical descriptions of diminished affective response, as participants described profound feelings of hopelessness and demoralization as well as high levels of anxiety and restlessness.

Overall, our results suggest that there may be many more experiential features that are relevant to understanding and assessing depersonalization than are typically elicited or discussed in research and clinical and non-clinical settings. In addition, they indicate potentially important targets for interventions for people with DP experiences. Our findings also point to a need for additional research to specify the nature of their relationship to the depersonalization range of experiences.

5 Limitations and future directions

Our study presents several limitations. Due to the self-report nature of these comments, crucial details that might affect the understanding of a participant's statement may not always have been present in the same way that might be elicited by an in-depth interview. Therefore, analysis erred on the side of caution when assigning codes and themes, and left out any statements that could be argued as representing a commonplace or typical form of experience (rather than assuming a more dramatic alteration). The use of self-reported scales is associated with other various limitations: (1) the answers are dependent on the honesty of participants; (2) the use of scales with subjective questions may difficult its understanding/interpretation and the introspective ability to provide an accurate answer;

and (3) its use may have skewed our population, as people prone to answering questionnaires, and therefore with different personal characteristics, are more likely to have answered it. We believe that the benefit of collecting data from a relatively large sample on a large number of potential experiences is worth the loss that comes from somewhat limited self-report comments. However, future research would benefit from conducting in-depth interviews on a similar range of topics with more participants.

Intriguingly, feelings of being a neutral observer of one's perceptions, emotions and actions has been reported in connexion with contemplative practices such as meditation (Lindahl & Britton 2019; Ciaunica et al., 2020). For example, in our study, the participants similarly described feelings of self-split or being an "external" observer: "Sometimes I feel like I'm a watching a movie of myself" (P218). Or "I also feel like the me who speaks is somewhat separate to the me who thinks" (P217). Future work needs to examine and to disentangle the positive versus negative valence of these feelings of self-detachment as reported in depersonalization and in meditation, as well as their underlying neural mechanisms and associated phenomenological markers.

In addition, this study cannot tell us whether the experiences reported here are due to (or cause) DP alone, but are merely descriptive, indicating other areas for clinical exploration and study.

The study excluded participants with any neurological conditions, and who were currently taking antidepressants or antipsychotic medications, effectively excluding major mental health conditions from the study. It is possible that participants did experience depression or anxiety (and indeed noted in some participants' descriptions), but such comorbid symptoms are fairly common in DPDR, and ostensibly in high levels of DP in the general population, so it would be impractical and unrealistic to exclude these from the study. This paper therefore cannot determine whether various experiences are "purely" related to DP alone, but whether they tend to occur in persons who experience high levels of DP.

Finally, further work needs to combine qualitative phenomenological tools, with quantitative behavioral and neural measures in order to provide an integrative examination of depersonalization experiences, providing thereby a much needed rich and multifaceted approach to this neglected yet widespread condition.

Notes

1. The interviews have been refined through clinical trials and with input from "experts by experience" (persons with schizophrenia and other psychoses) and clinicians familiar with phenomenological approaches to schizophrenia.

2. Validity and reliability trials are currently underway. The EAWE-SR may be accessed by contacting one of its authors (**Author). The items on the EAWE-SR are identical to those on the EAWE (Sass et al., 2017), though with different wording and response options.
3. For example, representationalists argue that while we typically have access only the representation's intentional content (something in the world which it's about) without noticing its non-intentional carrier properties (Harman, 1990; Moore, 1903; Tye, 1999), the process itself of constructing inner representations can become available to our introspective attention. Whenever we consciously direct our attention introspectively inwards, so to speak, the transparent processing of mental representations (typically taken for granted and hence "invisible") becomes "*opaque*", that is, "visible" and available to our attention (cf. the window metaphor described above).

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Notes on contributors

Dr. Anna Ciaunica is Principal Investigator at the Center for Philosophy of Science, University of Lisbon, Portugal, and Research Associate at the Institute of Cognitive Neuroscience, University College London, London, the UK. She worked as postdoctoral researcher at the Institute of Philosophy, Porto, Portugal and she was Research Associate at the Department of Clinical, Educational and Health Psychology, University College London. Before that she was postdoctoral researcher at the Department of Philosophy, University of Fribourg, Switzerland. Dr Ciaunica's work focuses on self-consciousness in humans and artificial agents. She is currently Leading Investigator of three interdisciplinary projects looking at the relationship between self-consciousness, embodiment and social interactions in humans and artificial agents. She is also the main coordinator of the Network for Embodied Consciousness, Technology and the Arts (NECTArts) – a collaborative platform bringing together artists, researchers, stakeholders, policy makers and people with lived experiences. NECTArts aims at fostering creative approaches to timely issues such as self-awareness and (dis)embodiment in our hyper-digitalized world.

Dr. Liz Pienkos is a licensed clinical psychologist and an assistant professor of psychology at Clarkson University in Potsdam, New York. She is the co-first-author (with Dr. Louis Sass) of the Examination of Anomalous World Experience. Her research focuses on the phenomenology of schizophrenia, using qualitative methods to explore the mechanisms and features of this and other psychiatric disorders.

Dr. Estelle Nakul is a postdoctoral researcher in cognitive neuroscience, formerly working at the CNRS and Aix-Marseille University, in France, now at the Ecole Polytechnique Fédérale de Lausanne, in Switzerland. She investigates the multisensory and cerebral mechanisms of bodily self-consciousness, with a focus on the contributions of the vestibular system.

Prof Luís Madeira develops his career in four synchronous strands. Psychiatrist, graduated in Medicine from the University of Lisbon in 2008, vice-president of Portuguese Society of Psychiatry, works at the North Lisbon Hospital Centre, Lusíadas Hospital and CUF Descobertas Hospital. He holds a Master's Degree in Philosophy from the University of Central Lancashire and a PhD in Medicine - in the field of Philosophy of Psychiatry. He is Assistant Professor of Medical Ethics and Deontology and of Psychiatry at the Faculty of Medicine of the University of Lisbon. He is also a Psychotherapist with the Portuguese Society for Client-Centred Psychotherapy and Person-Centred Approach.

Dr Harry Farmer is a Senior Lecturer in Psychology in the School of Human Sciences at the University of Greenwich. Harry obtained a BSc Psychology and Philosophy from the University of Bristol and then an MSc in Psychological Research from the University of Oxford before completing his PhD in Psychology at Royal Holloway, University of London in 2014. He worked as a Research Associate in Social Neuroscience at UCL's Institute of Cognitive Neuroscience between 2014 and 2018 and as a Research Associate in Virtual Reality at the University of Bath between 2017 and 2020. He joined the University of Greenwich in 2020. Harry's research interests centre on self-representation and social cognition. He uses a variety of methodologies including psychophysics, neuroimaging, physiological recordings and virtual reality.

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